Bachelor of Computer Applications

Semester – V

Business Analytics

Assignment No.: 6

Title: SalesPulse: Sorting & Filtering Data in Power BI to Identify High-Value Transactions and Underperforming Products

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SalesPulse: Sorting & Filtering Data in Power BI to Identify High-Value Transactions and Underperforming Products

1. Problem Statement

A sales manager wants to identify the company's **highest-value transactions** and its **underperforming products**. The raw sales ledger contains 300 transactions and cannot be read directly for these answers. The challenge is to use **Power BI's sorting and filtering techniques** (in Power Query Editor and in Report view) to extract the relevant business information quickly and reliably, so that the manager can protect and replicate what works, and review what does not.

2. Dataset Overview

The dataset (**sales_analysis_dataset_1.xlsx**) is a synthetic sales ledger for the financial year **October 2025 – September 2026**. The **Sales_Data** sheet holds **300 transactions** across **12 products**, **3 categories** (Hardware, Software, Services) and **4 regions** (North, South, East, West), with 14 columns:

Column	Description / Data Type
Transaction ID, Customer, Salesperson	Text – transaction number (TXN-1001 …), buyer and seller
Date	Date – date of sale
Region, Product, Category, Returned	Text – where, what and category; Returned = Y / N
Quantity	Whole Number – units sold
Unit Price ($), Unit Cost ($)	Decimal Number – selling price and cost per unit
Discount %	Percentage – discount given
Revenue ($)	Decimal – Quantity × Unit Price × (1 − Discount %)
Profit ($)	Decimal – Revenue − Quantity × Unit Cost

The dataset has **no missing values**, so no cleaning was needed. Total revenue is **$4,289,922** and total profit is **$1,706,242** (overall margin 39.8%).

3. Tools and Approach

The work was done in **Microsoft Power BI Desktop**. The Excel file was loaded using **Home → Get Data → Excel workbook**, and the **Sales_Data** table was opened in the **Power Query Editor** (Transform Data). Power Query was used because every transformation is recorded as an **Applied Step**, which makes the work transparent, repeatable, and easy to refresh. The original data is never changed. For each question a new query was created from Sales_Data (right-click → **Reference**), so that every result is a separate, clearly named query.

Two business definitions guide the analysis:

- **High-value transaction:** Revenue ≥ **$32,400** (about the 90th percentile of the dataset), which gives 32 of the 300 transactions.
- **Underperforming product:** total revenue below **60% of the average product revenue** (average $357,493.5, so the cut-off is **$214,496**) **or** profit margin below **15%**.

The approach has four parts: **Part A** – sorting transactions in Power Query; **Part B** – filtering transactions; **Part C** – grouping, sorting and filtering products to find underperformers; **Part D** – presenting the results in Report view with visual-level filters and sorting.

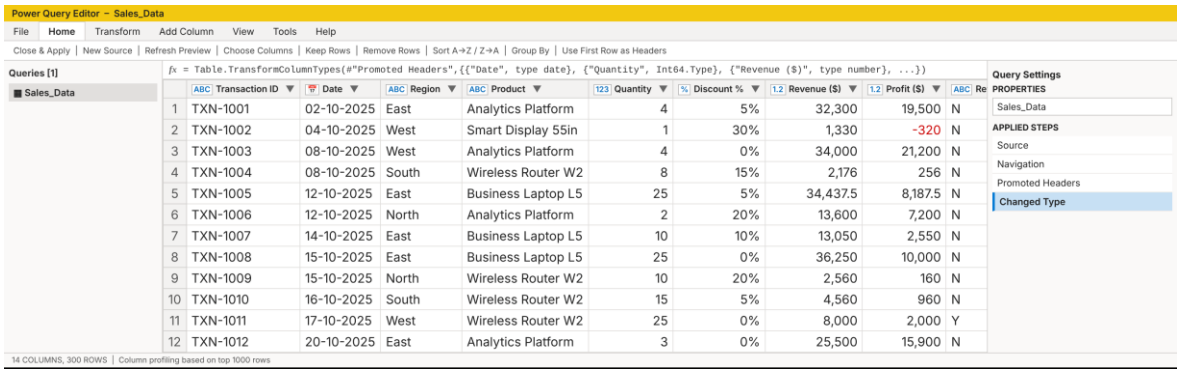
4. Step-by-Step Process

Part A: Loading and Sorting

Step 1: Loading the Data and Setting Data Types

Excel Feature Used: *Get Data → Excel workbook → Transform Data (Applied Steps: Source, Navigation, Promoted Headers, Changed Type)*

The workbook was imported and the Sales_Data table was selected in the Navigator. In Power Query Editor the first row was promoted to headers and data types were assigned: **Date** as Date, **Quantity** as Whole Number, prices, **Revenue** and **Profit** as Decimal Number, **Discount %** as Percentage and the rest as Text. The data is shown in its original date order, where the biggest deals are scattered through the year.



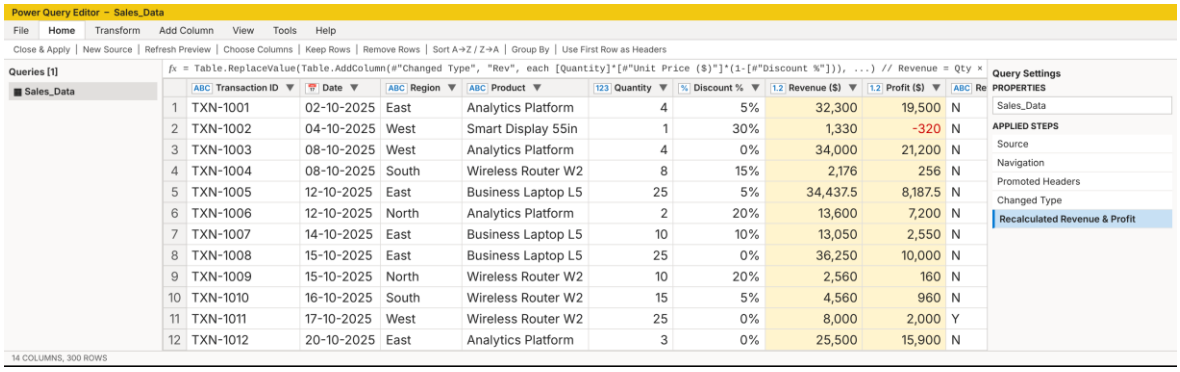
	Transaction ID	Date	Region	Product	Quantity	Discount %	Revenue ($)	Profit ($)	Re
1	TXN-1001	02-10-2025	East	Analytics Platform	4	5%	32,300	19,500	N
2	TXN-1002	04-10-2025	West	Smart Display 55in	1	30%	1,330	-320	N
3	TXN-1003	08-10-2025	West	Analytics Platform	4	0%	34,000	21,200	N
4	TXN-1004	08-10-2025	South	Wireless Router W2	8	15%	2,176	256	N
5	TXN-1005	12-10-2025	East	Business Laptop L5	25	5%	34,437.5	8,187.5	N
6	TXN-1006	12-10-2025	North	Analytics Platform	2	20%	13,600	7,200	N
7	TXN-1007	14-10-2025	East	Business Laptop L5	10	10%	13,050	2,550	N
8	TXN-1008	15-10-2025	East	Business Laptop L5	25	0%	36,250	10,000	N
9	TXN-1009	15-10-2025	North	Wireless Router W2	10	20%	2,560	160	N
10	TXN-1010	16-10-2025	South	Wireless Router W2	15	5%	4,560	960	N
11	TXN-1011	17-10-2025	West	Wireless Router W2	25	0%	8,000	2,000	Y
12	TXN-1012	20-10-2025	East	Analytics Platform	3	0%	25,500	15,900	N

Figure 1: Sales_Data loaded in Power Query Editor with Applied Steps

Step 2: Recalculating Revenue and Profit

Excel Feature Used: *Add Column → Custom Column (Applied Step: Recalculated Revenue & Profit)*

Revenue and Profit are stored as Excel formulas, so they were recalculated in Power Query with a Custom Column to guarantee that Power BI always has the values: **Revenue = Quantity × Unit Price × (1 − Discount %)** and **Profit = Revenue − Quantity × Unit Cost**. This also confirms that the numbers used in all later steps are consistent with the source.



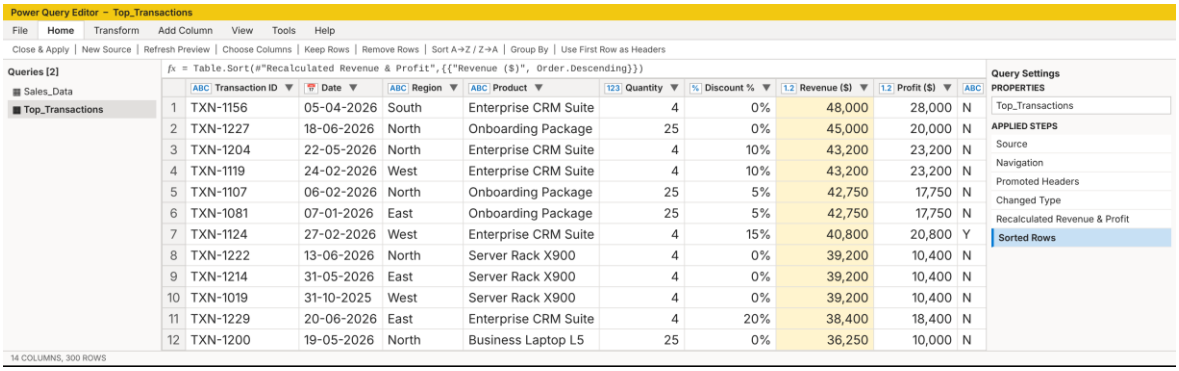
	Transaction ID	Date	Region	Product	Quantity	Discount %	Revenue ($)	Profit ($)	Re
1	TXN-1001	02-10-2025	East	Analytics Platform	4	5%	32,300	19,500	N
2	TXN-1002	04-10-2025	West	Smart Display 55in	1	30%	1,330	-320	N
3	TXN-1003	08-10-2025	West	Analytics Platform	4	0%	34,000	21,200	N
4	TXN-1004	08-10-2025	South	Wireless Router W2	8	15%	2,176	256	N
5	TXN-1005	12-10-2025	East	Business Laptop L5	25	5%	34,437.5	8,187.5	N
6	TXN-1006	12-10-2025	North	Analytics Platform	2	20%	13,600	7,200	N
7	TXN-1007	14-10-2025	East	Business Laptop L5	10	10%	13,050	2,550	N
8	TXN-1008	15-10-2025	East	Business Laptop L5	25	0%	36,250	10,000	N
9	TXN-1009	15-10-2025	North	Wireless Router W2	10	20%	2,560	160	N
10	TXN-1010	16-10-2025	South	Wireless Router W2	15	5%	4,560	960	N
11	TXN-1011	17-10-2025	West	Wireless Router W2	25	0%	8,000	2,000	Y
12	TXN-1012	20-10-2025	East	Analytics Platform	3	0%	25,500	15,900	N

Figure 2: Revenue ($) and Profit ($) recalculated using Custom Columns

Step 3: Sorting by Revenue (Largest to Smallest)

Excel Feature Used: *Right-click Revenue ($) → Sort Descending (Applied Step: Sorted Rows)*

A new query, **Top_Transactions**, was created and sorted on **Revenue ($)** in descending order. The largest transaction is **TXN-1156** (Enterprise CRM Suite, South, Juniper Foods) at **$48,000**, followed by **TXN-1227** (Onboarding Package, North) at $45,000, and two Enterprise CRM Suite deals at $43,200. The top 10 are dominated by Enterprise CRM Suite, Onboarding Package and Server Rack X900.



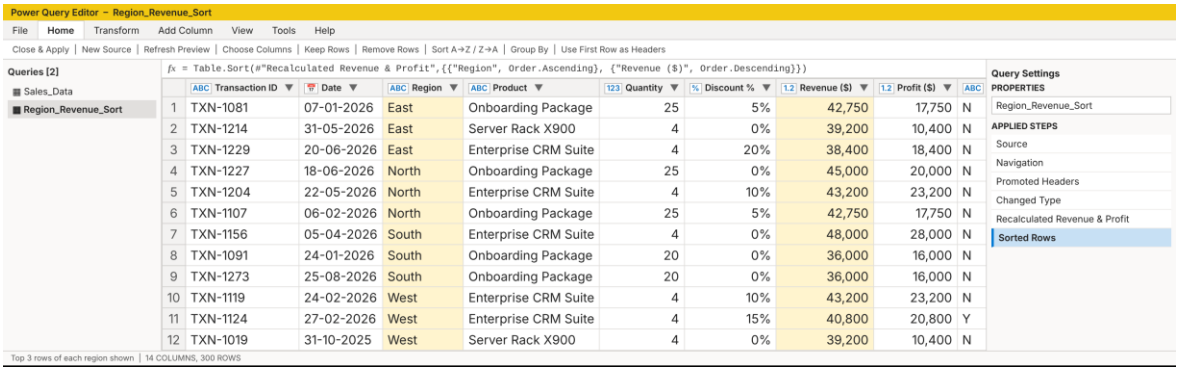
	Transaction ID	Date	Region	Product	Quantity	Discount %	Revenue ($)	Profit ($)	
1	TXN-1156	05-04-2026	South	Enterprise CRM Suite	4	0%	48,000	28,000	N
2	TXN-1227	18-06-2026	North	Onboarding Package	25	0%	45,000	20,000	N
3	TXN-1204	22-05-2026	North	Enterprise CRM Suite	4	10%	43,200	23,200	N
4	TXN-1119	24-02-2026	West	Enterprise CRM Suite	4	10%	43,200	23,200	N
5	TXN-1107	06-02-2026	North	Onboarding Package	25	5%	42,750	17,750	N
6	TXN-1081	07-01-2026	East	Onboarding Package	25	5%	42,750	17,750	N
7	TXN-1124	27-02-2026	West	Enterprise CRM Suite	4	15%	40,800	20,800	Y
8	TXN-1222	13-06-2026	North	Server Rack X900	4	0%	39,200	10,400	N
9	TXN-1214	31-05-2026	East	Server Rack X900	4	0%	39,200	10,400	N
10	TXN-1019	31-10-2025	West	Server Rack X900	4	0%	39,200	10,400	N
11	TXN-1229	20-06-2026	East	Enterprise CRM Suite	4	20%	38,400	18,400	N
12	TXN-1200	19-05-2026	North	Business Laptop L5	25	0%	36,250	10,000	N

Figure 3: Transactions sorted by Revenue ($) in descending order

Step 4: Multi-Level Sort: Region, then Revenue

Excel Feature Used: *Home → Sort Ascending on Region, then Sort Descending on Revenue (Sorted Rows)*

Sorting was applied on two columns: **Region (A to Z)** first and **Revenue (largest first)** second. Power Query records both in one Sorted Rows step, and the transactions are ranked inside each region. The best deal in each region is: **East** – TXN-1081 ($42,750), **North** – TXN-1227 ($45,000), **South** – TXN-1156 ($48,000) and **West** – TXN-1119 ($43,200). The figure shows the top 3 of each region.



	Transaction ID	Date	Region	Product	Quantity	Discount %	Revenue ($)	Profit ($)	
1	TXN-1081	07-01-2026	East	Onboarding Package	25	5%	42,750	17,750	N
2	TXN-1214	31-05-2026	East	Server Rack X900	4	0%	39,200	10,400	N
3	TXN-1229	20-06-2026	East	Enterprise CRM Suite	4	20%	38,400	18,400	N
4	TXN-1227	18-06-2026	North	Onboarding Package	25	0%	45,000	20,000	N
5	TXN-1204	22-05-2026	North	Enterprise CRM Suite	4	10%	43,200	23,200	N
6	TXN-1107	06-02-2026	North	Onboarding Package	25	5%	42,750	17,750	N
7	TXN-1156	05-04-2026	South	Enterprise CRM Suite	4	0%	48,000	28,000	N
8	TXN-1091	24-01-2026	South	Onboarding Package	20	0%	36,000	16,000	N
9	TXN-1273	25-08-2026	South	Onboarding Package	20	0%	36,000	16,000	N
10	TXN-1119	24-02-2026	West	Enterprise CRM Suite	4	10%	43,200	23,200	N
11	TXN-1124	27-02-2026	West	Enterprise CRM Suite	4	15%	40,800	20,800	Y
12	TXN-1019	31-10-2025	West	Server Rack X900	4	0%	39,200	10,400	N

Figure 4: Sorted by Region (A→Z) then Revenue (descending); top 3 per region shown

Step 5: Sorting by Profit (Smallest to Largest)

Excel Feature Used: *Right-click Profit ($) → Sort Ascending (Sorted Rows)*

High revenue does not always mean high profit, so a **Lowest_Profit** query was sorted on **Profit ($)** in ascending order, which brings the **worst** transactions to the top. Thirteen transactions have a **negative profit**, an early sign of which products are weak.

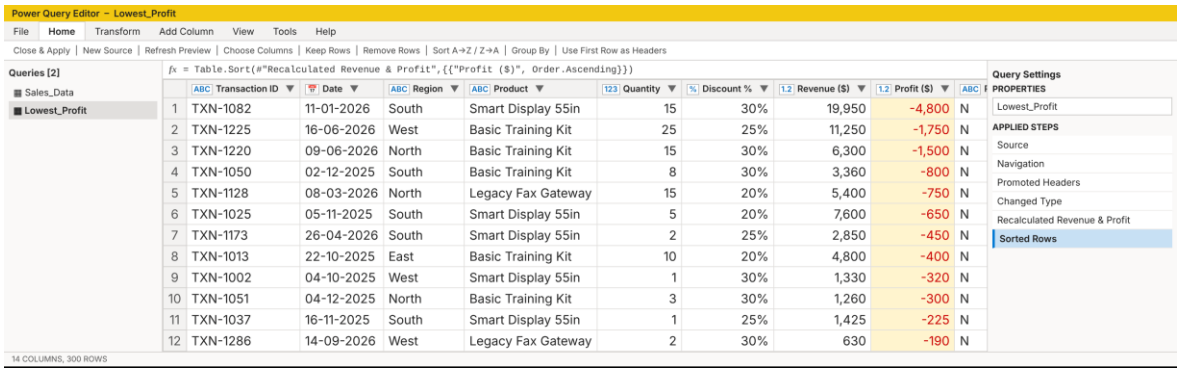


Figure 5: Transactions sorted by Profit ($) in ascending order

Part B: Filtering

Step 6: Filtering High-Value Transactions

Excel Feature Used: *Revenue ($) drop-down → Number Filters → Greater Than Or Equal To → 32400 (Filtered Rows)*

In the **High_Value** query, a Number Filter of **≥ 32,400** was applied on **Revenue ($)** and the result was sorted from largest to smallest. The status bar shows **32 rows**. These transactions (10.7% of all deals) bring in **$1,187,172.50**, which is **27.7% of total revenue**.

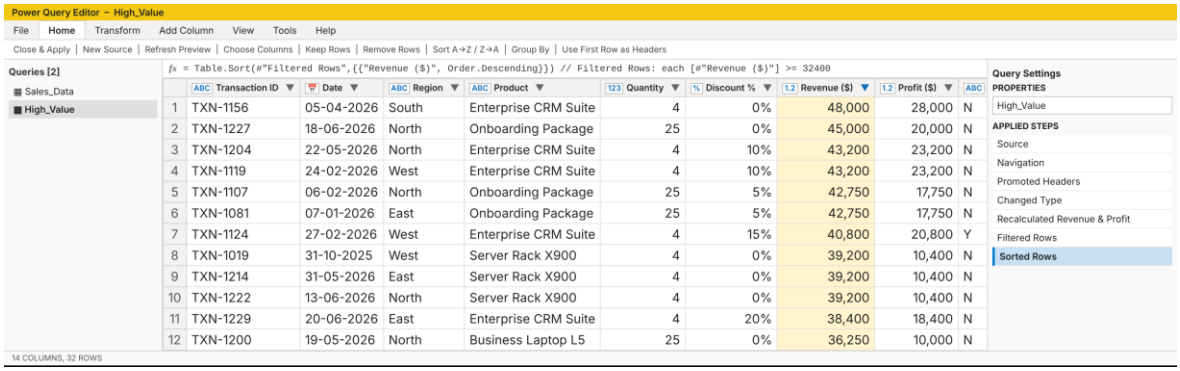


Figure 6: Number Filter on Revenue ($) ≥ 32,400 – 32 rows

Step 7: Combining Filters: Region and Category

Excel Feature Used: *Region drop-down → West; Category drop-down → Software (two Filtered Rows steps)*

Filters on different columns work together as an AND condition. Filtering **Region = West** and **Category = Software** returns **37 transactions** worth **$702,730**. This lets a manager look at one team and one product line together.

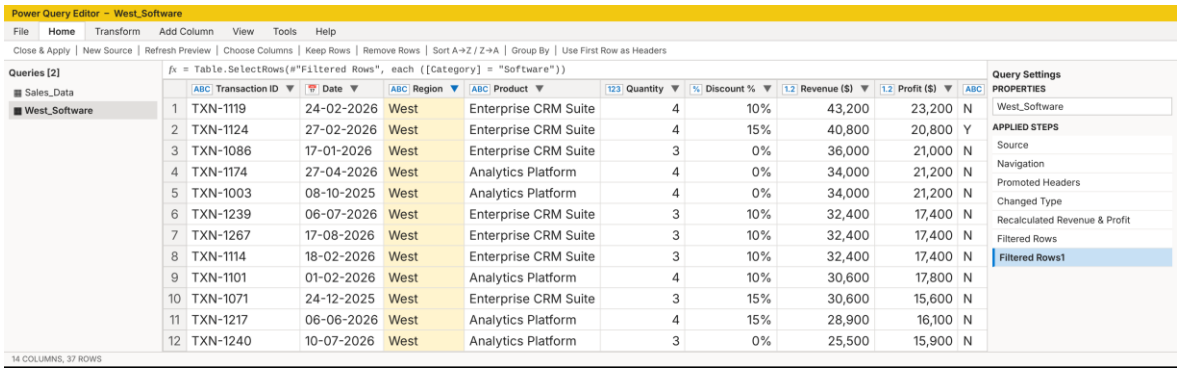


Figure 7: Region = West AND Category = Software – 37 rows

Step 8: Filtering Returned Sales

Excel Feature Used: *Returned drop-down → Text Filters → Equals → Y*

Filtering the **Returned** column for **Y** gives **10 transactions** with **$193,200** of returned revenue. This is a quality and customer-satisfaction check, because returns reduce the real value of high-revenue deals.

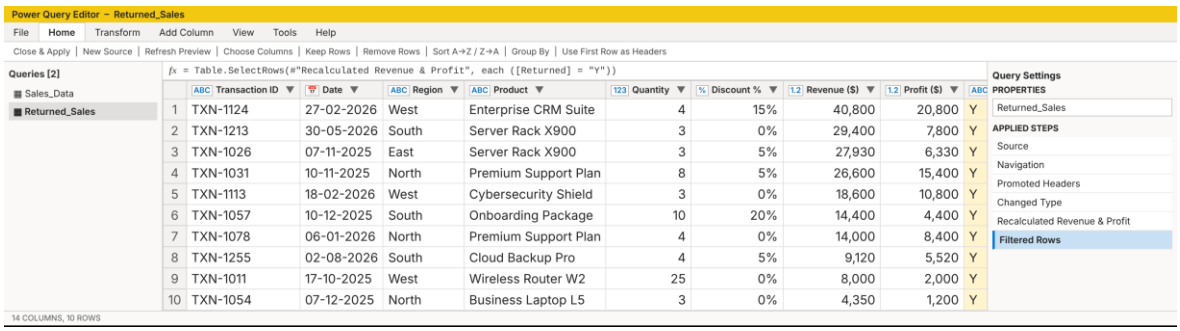


Figure 8: Returned = Y – 10 rows

Step 9: Filtering Heavily Discounted Sales

Excel Feature Used: *Discount % drop-down → Number Filters → Greater Than Or Equal To → 0.25*

A filter of **Discount % ≥ 25%** returns **9 transactions**. Together they make a **net loss of $10,335**, so very deep discounts destroy value. The manager may want an approval limit on discounts above 20%.

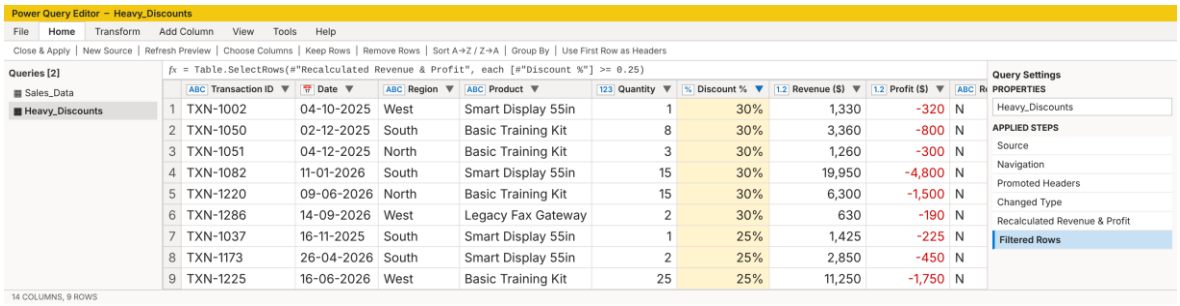
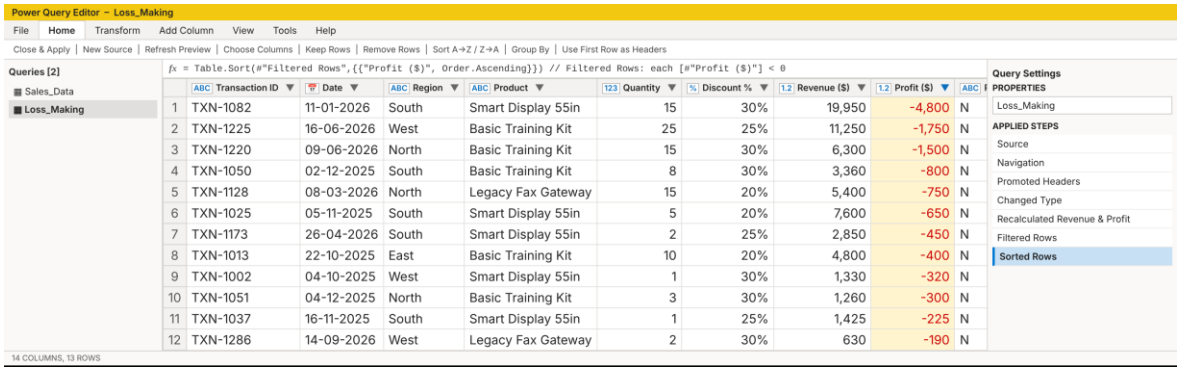


Figure 9: Discount % ≥ 25% – 9 rows

Step 10: Filtering Loss-Making Transactions

Excel Feature Used: *Profit ($) drop-down → Number Filters → Less Than → 0, then Sort Ascending*

Filtering **Profit ($) < 0** returns **13 transactions**, and all of them belong to just **three products**: **Smart Display 55in** (5), **Basic Training Kit** (5) and **Legacy Fax Gateway** (3). No other product ever sold at a loss.



	Transaction ID	Date	Region	Product	Quantity	Discount %	Revenue ($)	Profit ($)	
1	TXN-1082	11-01-2026	South	Smart Display 55in	15	30%	19,950	-4,800	N
2	TXN-1225	16-06-2026	West	Basic Training Kit	25	25%	11,250	-1,750	N
3	TXN-1220	09-06-2026	North	Basic Training Kit	15	30%	6,300	-1,500	N
4	TXN-1050	02-12-2025	South	Basic Training Kit	8	30%	3,360	-800	N
5	TXN-1128	08-03-2026	North	Legacy Fax Gateway	15	20%	5,400	-750	N
6	TXN-1025	05-11-2025	South	Smart Display 55in	5	20%	7,600	-650	N
7	TXN-1173	26-04-2026	South	Smart Display 55in	2	25%	2,850	-450	N
8	TXN-1013	22-10-2025	East	Basic Training Kit	10	20%	4,800	-400	N
9	TXN-1002	04-10-2025	West	Smart Display 55in	1	30%	1,330	-320	N
10	TXN-1051	04-12-2025	North	Basic Training Kit	3	30%	1,260	-300	N
11	TXN-1037	16-11-2025	South	Smart Display 55in	1	25%	1,425	-225	N
12	TXN-1286	14-09-2026	West	Legacy Fax Gateway	2	30%	630	-190	N

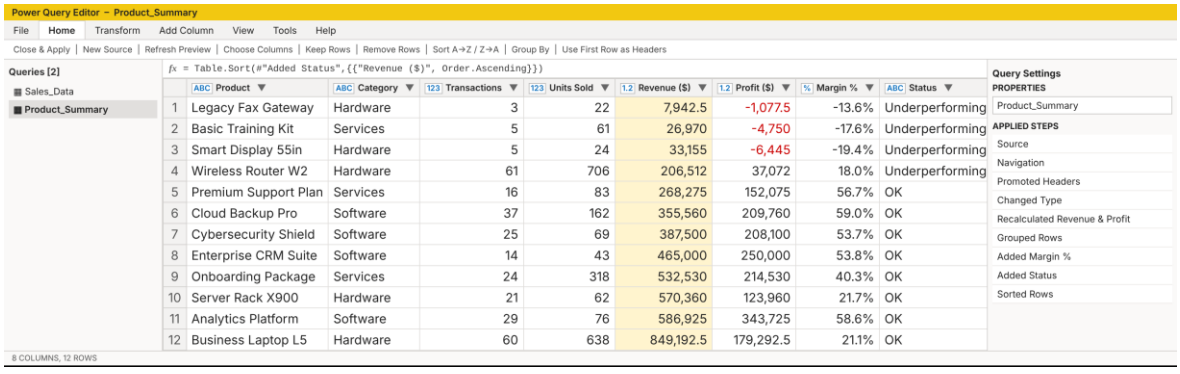
Figure 10: Profit ($) < 0 – 13 rows (first 12 shown)

Part C: Underperforming Products

Step 11: Grouping and Sorting Products by Revenue

Excel Feature Used: *Transform → Group By (Product) → Add Column → Custom Column → Sort Ascending*

A **Product_Summary** query was created with **Group By Product** (Transactions = Count, Units Sold, Revenue and Profit = Sum). **Margin % = Profit ÷ Revenue** and a **Status** column were added: a product is "Underperforming" if Revenue is below 60% of the average product revenue ($214,496) or Margin % is below 15%. Sorting by Revenue in ascending order puts the three lowest products first – **Legacy Fax Gateway ($7,942.5)**, **Basic Training Kit ($26,970)** and **Smart Display 55in ($33,155)** – each sold only 3 to 5 times and each loses money. The best product is **Business Laptop L5** with $849,192.5.



	Product	Category	Transactions	Units Sold	Revenue ($)	Profit ($)	Margin %	Status
1	Legacy Fax Gateway	Hardware	3	22	7,942.5	-1,077.5	-13.6%	Underperforming
2	Basic Training Kit	Services	5	61	26,970	-4,750	-17.6%	Underperforming
3	Smart Display 55in	Hardware	5	24	33,155	-6,445	-19.4%	Underperforming
4	Wireless Router W2	Hardware	61	706	206,512	37,072	18.0%	Underperforming
5	Premium Support Plan	Services	16	83	268,275	152,075	56.7%	OK
6	Cloud Backup Pro	Software	37	162	355,560	209,760	59.0%	OK
7	Cybersecurity Shield	Software	25	69	387,500	208,100	53.7%	OK
8	Enterprise CRM Suite	Software	14	43	465,000	250,000	53.8%	OK
9	Onboarding Package	Services	24	318	532,530	214,530	40.3%	OK
10	Server Rack X900	Hardware	21	62	570,360	123,960	21.7%	OK
11	Analytics Platform	Software	29	76	586,925	343,725	58.6%	OK
12	Business Laptop L5	Hardware	60	638	849,192.5	179,292.5	21.1%	OK

Figure 11: Product_Summary grouped by Product and sorted by Revenue ($) ascending

Step 12: Filtering the Underperforming Products

Excel Feature Used: *Status drop-down → Underperforming (Filtered Rows)*

Filtering **Status = Underperforming** returns **4 of 12 products**. Together they bring only **$274,579.5 (6.4%)** of total revenue.

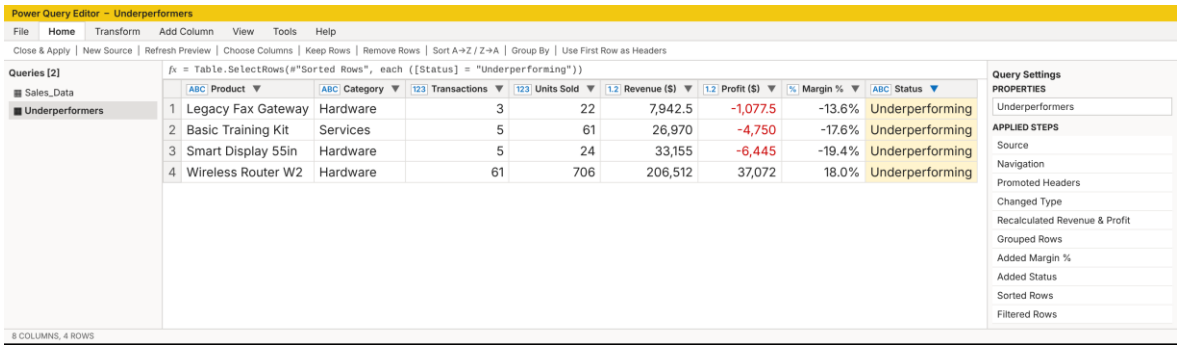


Figure 12: Status = Underperforming – 4 rows

Product	Revenue ($)	Margin %	Why flagged
Legacy Fax Gateway	7,942.50	−13.6%	Negative margin; revenue 97.8% below average
Basic Training Kit	26,970	−17.6%	Negative margin; revenue 92.5% below average
Smart Display 55in	33,155	−19.4%	Negative margin; revenue 90.7% below average
Wireless Router W2	206,512	18.0%	Revenue below the $214,496 cut-off; margin only 18% despite 706 units sold

Part D: Presenting in Report View

Step 13: Visual-Level Filter and Top N

Excel Feature Used: *Home → Close & Apply; Table visual → Filters pane → Revenue ($) ≥ 32400 and Top N (10)*

The queries were loaded with **Close & Apply**. In Report view a **Table** visual was placed with Transaction ID, Region, Product, Salesperson, Revenue and Profit. In the **Filters pane** the visual was filtered with **Revenue ($) is greater than or equal to 32400** and a second **Top N filter** (Top 10 by Revenue). Clicking the Revenue heading sorts the table, and two **Card** visuals show the **count (32)** and **total revenue ($1.19M)** of high-value transactions.

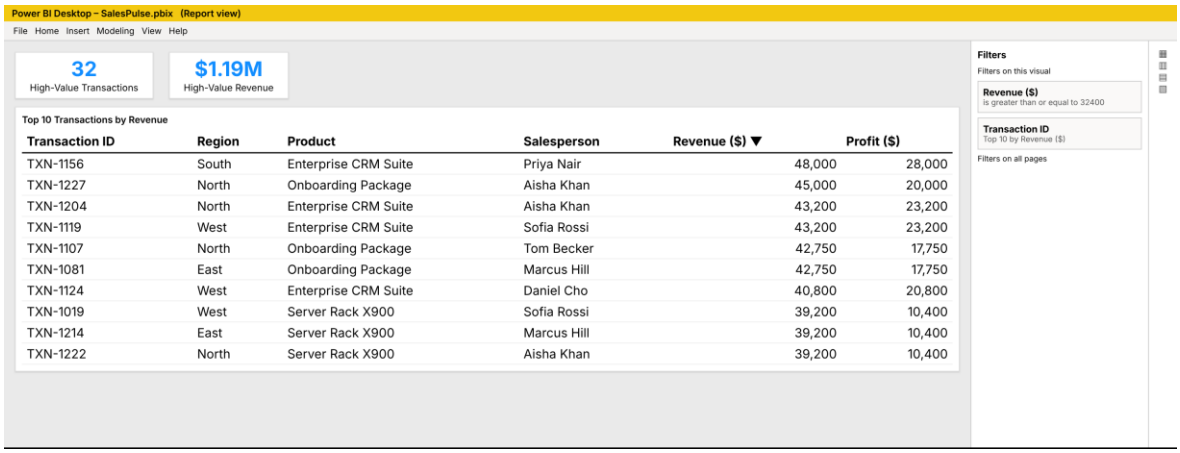


Figure 13: Report view – table visual with filters and cards for high-value transactions

Step 14: Sorting a Chart to Spot Underperformers

Excel Feature Used: *Clustered Bar Chart → ··· More options → Sort axis → Revenue ($) → Sort ascending*

A **Clustered Bar Chart** of **Revenue by Product** was added and sorted by Revenue in ascending order using the visual's **More options (···)** menu. Using **conditional colours**, the four underperforming products are shown in **red**. The chart makes the gap clear: the three loss-making products barely register against the best products.

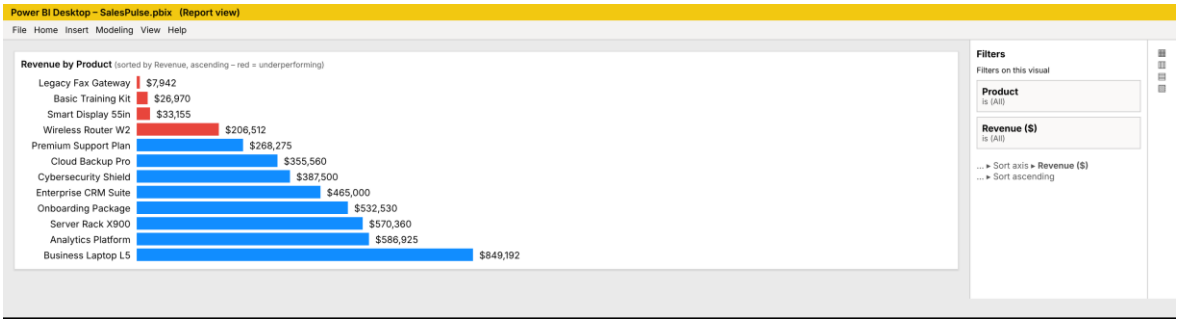


Figure 14: Revenue by Product sorted ascending; underperforming products in red

5. Key Business Findings

Question	Answer from Sorting / Filtering
Biggest single transaction	TXN-1156 – Enterprise CRM Suite, South, Juniper Foods, $48,000 (profit $28,000)
How many high-value deals (≥ $32,400)?	32 deals = $1,187,172.50 = 27.7% of revenue
Which region has most high-value deals?	West – 12 of 32; then North 8, South 6, East 6
Which products drive them?	Enterprise CRM Suite 10, Business Laptop L5 8, Onboarding Package 7, Server Rack X900 5, Analytics Platform 2
Top salesperson by high-value deals	Sofia Rossi (West) – 9 deals; then Aisha Khan 5, Marcus Hill 4
Loss-making transactions	13 transactions, all from Smart Display 55in, Basic Training Kit and Legacy Fax Gateway
Underperforming products	Legacy Fax Gateway, Basic Training Kit, Smart Display 55in, Wireless Router W2
Discount risk	9 deals at ≥ 25% discount lose $10,335 in total
Returns	10 returned sales worth $193,200

6. Data Quality Summary

The dataset required no cleaning: it has **no blanks, no duplicate IDs and consistent data types**. Revenue and Profit were recalculated in Power Query so they never depend on Excel's stored formula results. Because every sort, filter and calculation is a named Applied Step, the whole analysis **refreshes automatically** when the Excel file is updated. The thresholds ($32,400, 60% and 15%) are choices and can be changed in the filter steps or by using Power BI parameters.

7. Conclusion

Using **sorting and filtering in Power BI** (Power Query Editor and Report view), the 300-row sales ledger was turned into clear answers. The **highest-value transactions** are concentrated in a few products (**Enterprise CRM Suite, Business Laptop L5, Onboarding Package, Server Rack X900**) and in the **West** region, and 32 deals produce **27.7% of revenue**. The **underperforming products** are **Legacy Fax Gateway, Basic Training Kit, Smart Display 55in** (all loss-making) and **Wireless Router W2** (high volume but thin margin). The manager can therefore **protect and replicate the high-value deals**, **limit deep discounts** and **review, reprice or retire** the weak products.

8. Key Learnings

- **Sorting** ranks the data (best and worst first), while **filtering** isolates the records that meet a condition; the two are most powerful **together**.
- Power Query records every sort and filter as an **Applied Step**, so the analysis is **transparent, repeatable and refreshable**.
- A **multi-level sort** (Region, then Revenue) ranks records inside each group, and **Number / Text Filters** make business rules like "high-value" or "loss-making" easy to apply.
- Filters on **different columns combine as AND**, so questions like "West and Software" are answered in seconds.
- In Report view, the **Filters pane** (including **Top N**) and **Sort axis** let the manager explore results without changing the data.
- High revenue does not guarantee high profit – **checking profit and margin** is needed to find truly underperforming products.